BÁO CÁO

todo_app.py


```
35      def them_cong_viec():
36          noi_dung = bien_cong_viec_moi.get().strip()
37          if not noi_dung:
38              messagebox.showerror(
39                  "Loi", "Vui long nhap noi dung cong viec truoc khi them."
40              )
41              entry_cong_viec.focus_set()
42              return
43
44          danh_sach_cong_viec.append((noi_dung, False))
45          bien_cong_viec_moi.set("")
46          lam_moi_listbox()
47          entry_cong_viec.focus_set()
48
49      def lay_vi_tri_da_chon(thong_bao):
50          lua_chon = listbox_cong_viec.curselection()
51          if not lua_chon:
52              messagebox.showerror("Loi", thong_bao)
53              return None
54          return lua_chon[0]
55
56      def xoa_cong_viec():
57          vi_tri = lay_vi_tri_da_chon(
58              "Vui long chon mot cong viec trong danh sach de xoa."
59          )
60          if vi_tri is None:
61              return
62          danh_sach_cong_viec.pop(vi_tri)
63          lam_moi_listbox()
64
65      def danh_dau_hoan_thanh():
66          vi_tri = lay_vi_tri_da_chon(
67              "Vui long chon mot cong viec de danh dau hoan thanh."
68          )
```

todo_app.py

Buoi09 / Scripts / Trang 1/3

```python
import tkinter as tk
from tkinter import messagebox


def tao_giao_dien(cua_so):
    danh_sach_cong_viec = []

    cua_so.title("Ung dung Ghi chu cong viec")
    cua_so.geometry("440x410")
    cua_so.resizable(False, False)

    tk.Label(
        cua_so, text="TO-DO LIST", font=("Arial", 16, "bold")
    ).grid(row=0, column=0, columnspan=3, pady=10)

    bien_cong_viec_moi = tk.StringVar()
    entry_cong_viec = tk.Entry(
        cua_so, textvariable=bien_cong_viec_moi, width=36
    )
    entry_cong_viec.grid(row=1, column=0, columnspan=2, padx=10, pady=5)

    listbox_cong_viec = tk.Listbox(
        cua_so, width=52, height=13, exportselection=False
    )
    listbox_cong_viec.grid(
        row=2, column=0, columnspan=3, padx=10, pady=10
    )

    def lam_moi_listbox():
        listbox_cong_viec.delete(0, tk.END)
        for noi_dung, da_hoan_thanh in danh_sach_cong_viec:
            tien_to = "[Da xong] " if da_hoan_thanh else "[Chua xong] "
            listbox_cong_viec.insert(tk.END, tien_to + noi_dung)

```

todo_app.py | 1-34 / 96 dong

scale_demo.py

Buoi09 / Scripts / Ma nguon Python

```python
import tkinter as tk


def tao_giao_dien(cua_so):
    cua_so.title("Vi du Scale")
    cua_so.geometry("360x180")
    cua_so.resizable(False, False)

    nhan_gia_tri = tk.Label(cua_so, text="Gia tri hien tai: 0")
    nhan_gia_tri.pack(pady=(20, 8))

    def cap_nhat_gia_tri(gia_tri):
        nhan_gia_tri.config(text=f"Gia tri hien tai: {int(float(gia_tri))}")

    thanh_truot = tk.Scale(
        cua_so,
        from_=0,
        to=100,
        orient="horizontal",
        command=cap_nhat_gia_tri,
    )
    thanh_truot.pack(padx=20, fill="x")


if __name__ == "__main__":
    cua_so = tk.Tk()
    tao_giao_dien(cua_so)
    cua_so.mainloop()
```

scale_demo.py | 1-28 / 28 dong

may_tinh_bo_tui.py

Buoi09 / Scripts / Trang 4/4

```python

if __name__ == "__main__":
    cua_so = tk.Tk()
    tao_giao_dien(cua_so)
    cua_so.mainloop()
```

may_tinh_bo_tui.py | 103-107 / 107 dong

may_tinh_bo_tui.py

Buoi09 / Scripts / Trang 3/4

```python
        try:
            ket_qua = tinh_bieu_thuc(bieu_thuc)
        except (SyntaxError, ValueError, ZeroDivisionError):
            messagebox.showerror("Loi", "Bieu thuc khong hop le.")
            return
        bien_hien_thi.set(str(ket_qua))

    cac_nut = (
        ("7", 1, 0), ("8", 1, 1), ("9", 1, 2), ("/", 1, 3),
        ("4", 2, 0), ("5", 2, 1), ("6", 2, 2), ("*", 2, 3),
        ("1", 3, 0), ("2", 3, 1), ("3", 3, 2), ("-", 3, 3),
        ("0", 4, 0), (".", 4, 1), ("=", 4, 2), ("+", 4, 3),
    )
    for ky_tu, hang, cot in cac_nut:
        lenh = tinh if ky_tu == "=" else lambda k=ky_tu: them_vao_bieu_thuc(k)
        tk.Button(
            cua_so,
            text=ky_tu,
            font=("Arial", 14),
            command=lenh,
        ).grid(row=hang, column=cot, padx=4, pady=4, sticky="nsew")

    tk.Button(cua_so, text="C", command=xoa).grid(
        row=5, column=0, columnspan=2, padx=4, pady=4, sticky="nsew"
    )
    tk.Button(cua_so, text="Xoa ky tu", command=xoa_ky_tu_cuoi).grid(
        row=5, column=2, columnspan=2, padx=4, pady=4, sticky="nsew"
    )

    for cot in range(4):
        cua_so.grid_columnconfigure(cot, weight=1)
    for hang in range(1, 6):
        cua_so.grid_rowconfigure(hang, weight=1)

```

may_tinh_bo_tui.py | 69-102 / 107 dong

may_tinh_bo_tui.py


```
    return tinh_nut(cay)


def tao_giao_dien(cua_so):
    cua_so.title("May tinh bo tui")
    cua_so.geometry("300x390")
    cua_so.resizable(False, False)

    bien_hien_thi = tk.StringVar()
    o_hien_thi = tk.Entry(
        cua_so,
        textvariable=bien_hien_thi,
        justify="right",
        font=("Arial", 20),
        state="readonly",
        readonlybackground="white",
    )
    o_hien_thi.grid(
        row=0, column=0, columnspan=4, padx=10, pady=12, sticky="nsew"
    )

    def them_vao_bieu_thuc(ky_tu):
        bien_hien_thi.set(bien_hien_thi.get() + ky_tu)

    def xoa():
        bien_hien_thi.set("")

    def xoa_ky_tu_cuoi():
        bien_hien_thi.set(bien_hien_thi.get()[:-1])

    def tinh():
        bieu_thuc = bien_hien_thi.get().strip()
        if not bieu_thuc:
            return
```

may_tinh_bo_tui.py

Buoi09 / Scripts / Trang 1/4

```python
import ast
import operator
import tkinter as tk
from tkinter import messagebox


_PHEP_TOAN = {
    ast.Add: operator.add,
    ast.Sub: operator.sub,
    ast.Mult: operator.mul,
    ast.Div: operator.truediv,
}


def tinh_bieu_thuc(bieu_thuc):
    """Tinh bieu thuc so hoc chi voi +, -, *, / va khong dung eval()."""
    cay = ast.parse(bieu_thuc, mode="eval")

    def tinh_nut(nut):
        if isinstance(nut, ast.Expression):
            return tinh_nut(nut.body)
        if isinstance(nut, ast.Constant) and type(nut.value) in (int, float):
            return nut.value
        if isinstance(nut, ast.UnaryOp) and isinstance(
            nut.op, (ast.UAdd, ast.USub)
        ):
            gia_tri = tinh_nut(nut.operand)
            return gia_tri if isinstance(nut.op, ast.UAdd) else -gia_tri
        if isinstance(nut, ast.BinOp) and type(nut.op) in _PHEP_TOAN:
            return _PHEP_TOAN[type(nut.op)](
                tinh_nut(nut.left), tinh_nut(nut.right)
            )
        raise ValueError("Bieu thuc chi ho tro so va cac phep +, -, *, /.")

```

may_tinh_bo_tui.py | 1-34 / 107 dong

listbox_combobox.py

Buoi09 / Scripts / Trang 2/2

```python
    thanh_pho = ttk.Combobox(
        cua_so,
        values=("Ha Noi", "Da Nang", "TP.HCM", "Hue"),
        state="readonly",
    )
    thanh_pho.current(0)
    thanh_pho.pack(pady=4)

    ket_qua_thanh_pho = tk.Label(cua_so, text="")

    def xem_thanh_pho():
        ket_qua_thanh_pho.config(text=f"Ban chon: {thanh_pho.get()}")

    tk.Button(
        cua_so, text="Xac nhan", command=xem_thanh_pho
    ).pack(pady=4)
    ket_qua_thanh_pho.pack(pady=4)


if __name__ == "__main__":
    cua_so = tk.Tk()
    tao_giao_dien(cua_so)
    cua_so.mainloop()
```

listbox_combobox.py | 35-57 / 57 dong

listbox_combobox.py

Buoi09 / Scripts / Trang 1/2

```python
import tkinter as tk
from tkinter import ttk


def tao_giao_dien(cua_so):
    cua_so.title("Listbox va Combobox")
    cua_so.geometry("360x380")
    cua_so.resizable(False, False)

    tk.Label(cua_so, text="Bai tap 3.1 - Chon mon hoc").pack(pady=(12, 4))
    listbox_mon_hoc = tk.Listbox(cua_so, height=5, exportselection=False)
    for mon_hoc in ("Toan", "Ly", "Hoa", "Van", "Anh"):
        listbox_mon_hoc.insert(tk.END, mon_hoc)
    listbox_mon_hoc.pack(pady=4)

    ket_qua_mon_hoc = tk.Label(cua_so, text="")
    ket_qua_mon_hoc.pack(pady=4)

    def xem_mon_da_chon():
        lua_chon = listbox_mon_hoc.curselection()
        if not lua_chon:
            ket_qua_mon_hoc.config(text="Vui long chon mot mon hoc.")
            return
        ket_qua_mon_hoc.config(
            text=f"Mon da chon: {listbox_mon_hoc.get(lua_chon[0])}"
        )

    tk.Button(
        cua_so, text="Xem mon da chon", command=xem_mon_da_chon
    ).pack(pady=4)

    ttk.Separator(cua_so, orient="horizontal").pack(fill="x", padx=15, pady=10)

    tk.Label(cua_so, text="Bai tap 3.2 - Chon thanh pho").pack(pady=4)
```

listbox_combobox.py | 1-34 / 57 dong

form_validate_messagebox.py

Buoi09 / Scripts / Trang 2/2

```python
            entry_ho_ten.focus_set()
            return

        try:
            tuoi = int(tuoi_nhap)
        except ValueError:
            messagebox.showerror("Loi", "Tuoi phai la mot so nguyen.")
            entry_tuoi.focus_set()
            return

        if not email or "@" not in email:
            messagebox.showerror("Loi", "Email khong hop le (thieu ky tu @).")
            entry_email.focus_set()
            return

        messagebox.showinfo(
            "Thanh cong",
            f"Da nhan thong tin: {ho_ten} - {tuoi} tuoi - {email}",
        )

    tk.Button(
        cua_so, text="Submit", command=xu_ly_submit
    ).grid(row=3, column=0, columnspan=2, pady=15)


if __name__ == "__main__":
    cua_so = tk.Tk()
    tao_giao_dien(cua_so)
    cua_so.mainloop()
```

form_validate_messagebox.py | 35-63 / 63 dong

form_validate_messagebox.py

Buoi09 / Scripts / Trang 1/2

```python
import tkinter as tk
from tkinter import messagebox


def tao_giao_dien(cua_so):
    cua_so.title("Form co validate")
    cua_so.geometry("400x240")
    cua_so.resizable(False, False)

    tk.Label(cua_so, text="Ho ten:").grid(
        row=0, column=0, padx=10, pady=10, sticky="w"
    )
    entry_ho_ten = tk.Entry(cua_so, width=28)
    entry_ho_ten.grid(row=0, column=1, padx=10, pady=10)

    tk.Label(cua_so, text="Tuoi:").grid(
        row=1, column=0, padx=10, pady=10, sticky="w"
    )
    entry_tuoi = tk.Entry(cua_so, width=28)
    entry_tuoi.grid(row=1, column=1, padx=10, pady=10)

    tk.Label(cua_so, text="Email:").grid(
        row=2, column=0, padx=10, pady=10, sticky="w"
    )
    entry_email = tk.Entry(cua_so, width=28)
    entry_email.grid(row=2, column=1, padx=10, pady=10)

    def xu_ly_submit():
        ho_ten = entry_ho_ten.get().strip()
        tuoi_nhap = entry_tuoi.get().strip()
        email = entry_email.get().strip()

        if not ho_ten:
            messagebox.showerror("Loi", "Ho ten khong duoc de trong.")
```

form_validate_messagebox.py | 1-34 / 63 dong

checkbutton_radiobutton.py

Buoi09 / Scripts / Trang 3/3

```python
        tab_tai_khoan, text="Xac nhan", command=xem_tai_khoan
    ).pack(anchor="w", pady=8)
    ket_qua_tai_khoan.pack(anchor="w")


if __name__ == "__main__":
    cua_so = tk.Tk()
    tao_giao_dien(cua_so)
    cua_so.mainloop()
```

checkbutton_radiobutton.py | 69-77 / 77 dong

checkbutton_radiobutton.py


```
            ten
            for ten, bien in (
                ("Doc sach", doc_sach),
                ("The thao", the_thao),
                ("Am nhac", am_nhac),
            )
            if bien.get()
        ]
        ket_qua_so_thich.config(
            text="Ban da chon: " + (", ".join(da_chon) if da_chon else "Chua chon")
        )

    tk.Button(
        tab_so_thich, text="Xem lua chon", command=xem_so_thich
    ).pack(anchor="w", pady=4)

    loai_tai_khoan = tk.StringVar(value="Ca nhan")
    tk.Label(tab_tai_khoan, text="Chon loai tai khoan:").pack(
        anchor="w", pady=(0, 8)
    )
    for ten in ("Ca nhan", "Doanh nghiep"):
        tk.Radiobutton(
            tab_tai_khoan,
            text=ten,
            value=ten,
            variable=loai_tai_khoan,
        ).pack(anchor="w")

    ket_qua_tai_khoan = tk.Label(tab_tai_khoan, text="")

    def xem_tai_khoan():
        ket_qua_tai_khoan.config(text=f"Ban da chon: {loai_tai_khoan.get()}")

    tk.Button(
```

checkbutton_radiobutton.py

Buoi09 / Scripts / Trang 1/3

```python
import tkinter as tk
from tkinter import ttk


def tao_giao_dien(cua_so):
    cua_so.title("Checkbutton va Radiobutton")
    cua_so.geometry("360x300")
    cua_so.resizable(False, False)

    cac_tab = ttk.Notebook(cua_so)
    tab_so_thich = ttk.Frame(cac_tab, padding=12)
    tab_tai_khoan = ttk.Frame(cac_tab, padding=12)
    cac_tab.add(tab_so_thich, text="So thich")
    cac_tab.add(tab_tai_khoan, text="Loai tai khoan")
    cac_tab.pack(fill="both", expand=True)

    doc_sach = tk.BooleanVar(value=False)
    the_thao = tk.BooleanVar(value=False)
    am_nhac = tk.BooleanVar(value=False)
    tk.Label(tab_so_thich, text="Chon so thich cua ban:").pack(
        anchor="w", pady=(0, 8)
    )
    for ten, bien in (
        ("Doc sach", doc_sach),
        ("The thao", the_thao),
        ("Am nhac", am_nhac),
    ):
        tk.Checkbutton(tab_so_thich, text=ten, variable=bien).pack(anchor="w")

    ket_qua_so_thich = tk.Label(tab_so_thich, text="", wraplength=310)
    ket_qua_so_thich.pack(anchor="w", pady=8)

    def xem_so_thich():
        da_chon = [
```

checkbutton_radiobutton.py | 1-34 / 77 dong

bien_dieu_khien.py

Buoi09 / Scripts / Trang 2/2

```python

    def xem_gia_tri():
        ket_qua.config(
            text=(
                f"Ho ten: {bien_ho_ten.get()}\n"
                f"Tuoi: {bien_tuoi.get()}\n"
                f"Dong y dieu khoan: {bien_dong_y.get()}"
            )
        )
        bien_tuoi.set(bien_tuoi.get() + 1)

    tk.Button(
        cua_so, text="Xem gia tri (va +1 tuoi)", command=xem_gia_tri
    ).grid(row=3, column=0, columnspan=2, pady=5)


if __name__ == "__main__":
    cua_so = tk.Tk()
    tao_giao_dien(cua_so)
    cua_so.mainloop()

# textvariable tao lien ket hai chieu giua Entry va bien Tkinter: .get() doc
# gia tri moi nhat, con .set() co the cap nhat Entry. Entry.get() chi doc truc
# tiep mot widget, khong tien neu can chia se hoac cap nhat gia tri qua bien.
```

bien_dieu_khien.py | 35-58 / 58 dong

bien_dieu_khien.py

Buoi09 / Scripts / Trang 1/2

```python
import tkinter as tk


def tao_giao_dien(cua_so):
    cua_so.title("Vi du bien dieu khien")
    cua_so.geometry("380x230")
    cua_so.resizable(False, False)

    bien_ho_ten = tk.StringVar(value="Nguyen Van A")
    bien_tuoi = tk.IntVar(value=18)
    bien_dong_y = tk.BooleanVar(value=False)

    tk.Label(cua_so, text="Ho ten:").grid(
        row=0, column=0, padx=10, pady=10, sticky="w"
    )
    tk.Entry(cua_so, textvariable=bien_ho_ten, width=25).grid(
        row=0, column=1, padx=10, pady=10
    )

    tk.Label(cua_so, text="Tuoi:").grid(
        row=1, column=0, padx=10, pady=10, sticky="w"
    )
    tk.Entry(cua_so, textvariable=bien_tuoi, width=25).grid(
        row=1, column=1, padx=10, pady=10
    )

    tk.Checkbutton(
        cua_so,
        text="Dong y dieu khoan",
        variable=bien_dong_y,
    ).grid(row=2, column=0, columnspan=2, pady=5)

    ket_qua = tk.Label(cua_so, text="", justify="left")
    ket_qua.grid(row=4, column=0, columnspan=2, padx=10, pady=5, sticky="w")
```

bien_dieu_khien.py | 1-34 / 58 dong

todo_app.py

Buoi09 / Scripts / Trang 3/3

```python
        if vi_tri is None:
            return
        noi_dung, _ = danh_sach_cong_viec[vi_tri]
        danh_sach_cong_viec[vi_tri] = (noi_dung, True)
        lam_moi_listbox()
        listbox_cong_viec.selection_set(vi_tri)

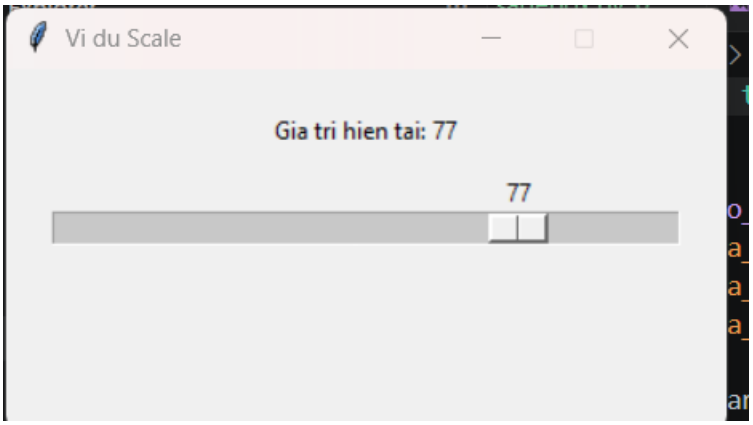
    tk.Button(
        cua_so, text="Them", command=them_cong_viec
    ).grid(row=1, column=2, padx=10, pady=5)
    tk.Button(
        cua_so, text="Danh dau hoan thanh", command=danh_dau_hoan_thanh
    ).grid(row=3, column=0, padx=8, pady=5)
    tk.Button(
        cua_so, text="Xoa cong viec", command=xoa_cong_viec
    ).grid(row=3, column=1, padx=8, pady=5)

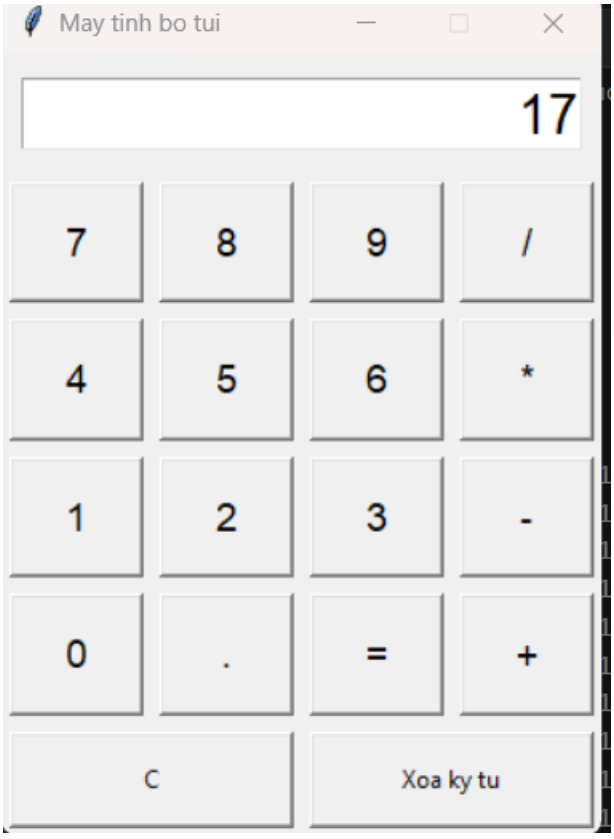
    def them_khi_nhan_enter(event):
        if event.keysym == "Return":
            them_cong_viec()

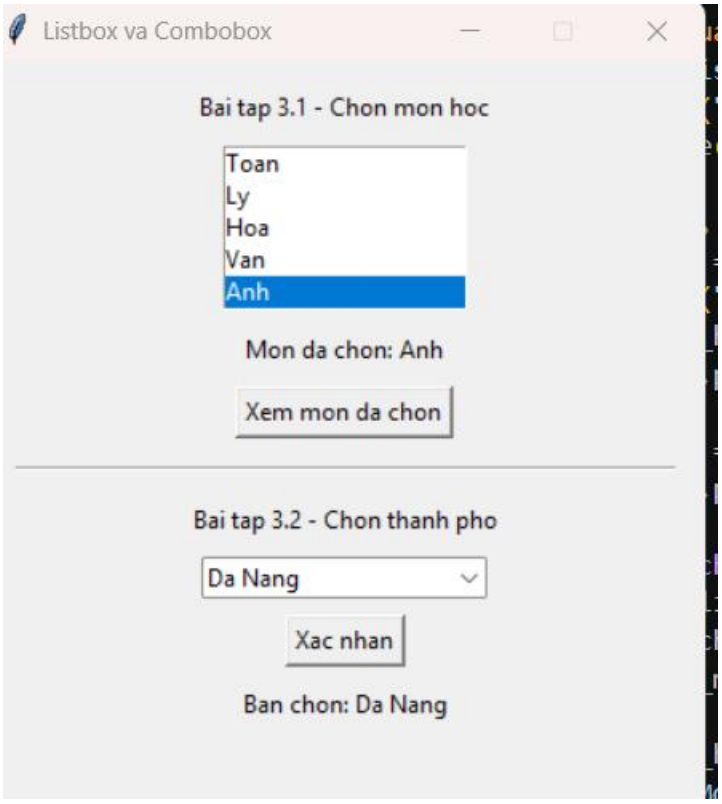
    entry_cong_viec.bind("<Return>", them_khi_nhan_enter)

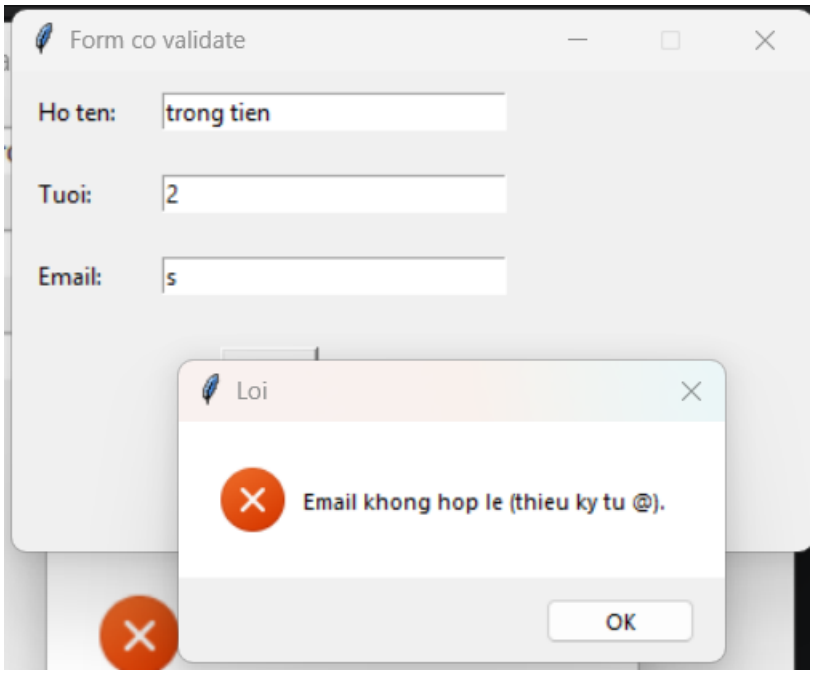

if __name__ == "__main__":
    cua_so = tk.Tk()
    tao_giao_dien(cua_so)
    cua_so.mainloop()
```

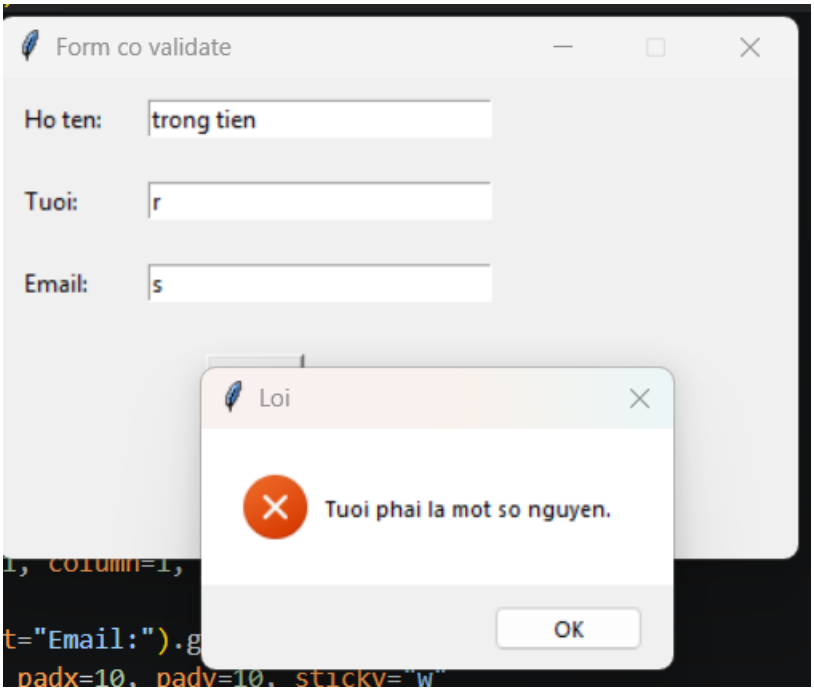
todo_app.py | 69-96 / 96 dong

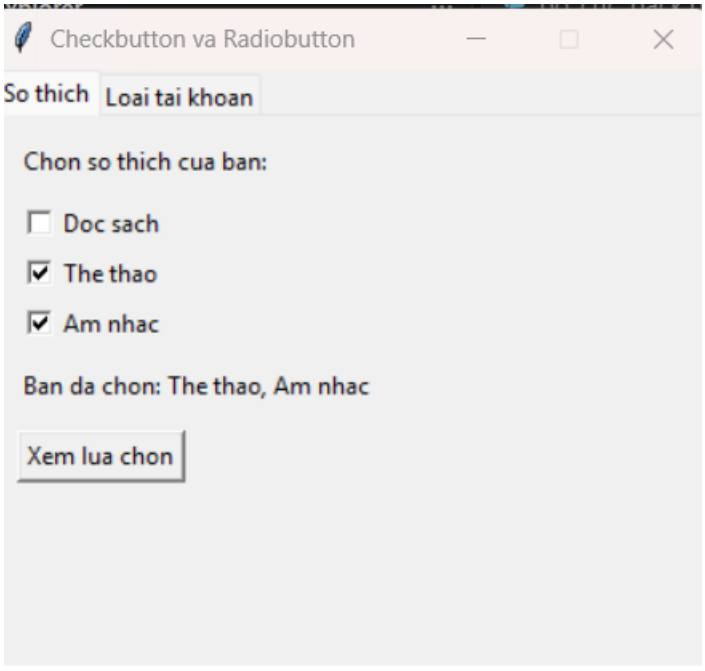

Vi du Scale
Gia tri hien tai: 77
77

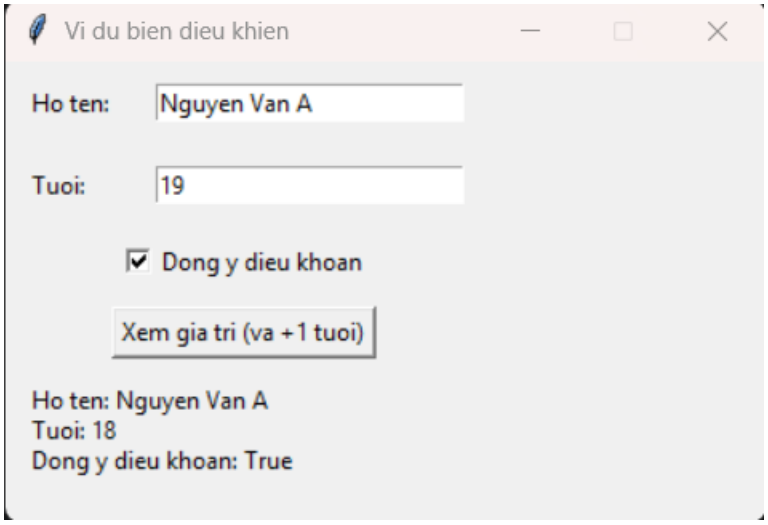

May tinh bo tui
17
7
8
9
/
4
5
6
*
1
2
3
-
0
.
=
+
C
Xoa ky tu

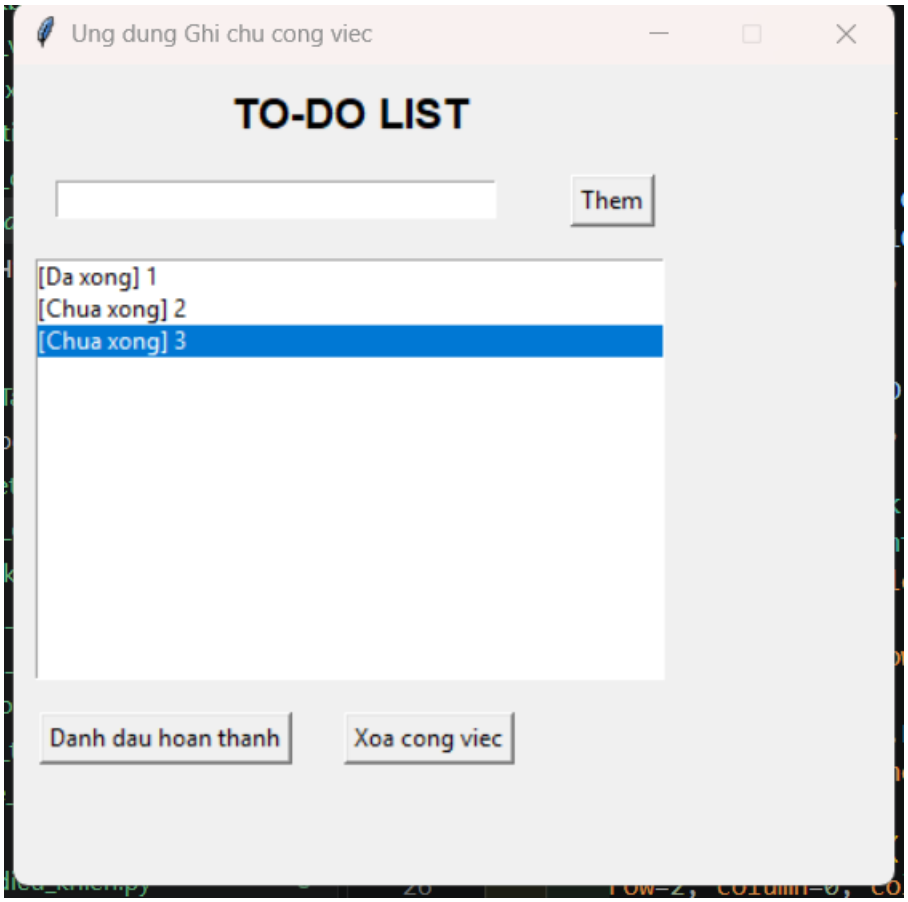

Listbox va Combobox
Bai tap 3.1 - Chon mon hoc
Toan
Ly
Hoa
Van
Anh
Mon da chon: Anh
Xem mon da chon
Bai tap 3.2 - Chon thanh pho
Da Nang
Xac nhan
Ban chon: Da Nang


Form co validate
Ho ten:
trong tien
Tuoi:
2
Email:
s
Loi
Email khong hop le (thieu ky tu @).
OK


Form co validate
Ho ten:
trong tien
Tuoi:
r
Email:
s
Loi
Tuoi phai la mot so nguyen.
OK
t="Email:").g
padx=10, pady=10, sticky="w"


Checkbutton va Radiobutton
So thich
Loai tai khoan
Chon so thich cua ban:
Doc sach
The thao
Am nhac
Ban da chon: The thao, Am nhac
Xem lua chon


Vi du bien dieu khien
Ho ten:
Nguyen Van A
Tuoi:
19
Dong y dieu khoan
Xem gia tri (va +1 tuoi)
Ho ten: Nguyen Van A
Tuoi: 18
Dong y dieu khoan: True


Ung dung Ghi chu cong viec
TO-DO LIST
Them
[Da xong] 1
[Chua xong] 2
[Chua xong] 3
Danh dau hoan thanh
Xoa cong viec